



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 -65



**KOLBI Cassava Grater
Model: KG500**

KOLBI KG500 CASSAVA GRATER

Test Report No.	:	2013 - 65
Date of Test	:	May 11, 2013
Test Valid Until	:	May 11, 2018
Brand	:	KOLBI
Model	:	KG500
Type	:	Portable, circumferential-type
Manufacturer	:	TROPICS AGRO INDUSTRIES, INC. 25 Panganiban Drive Naga City
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The KOLBI KG500 Cassava Grater was tested using peeled cassava with an average Moisture Content of 54.2%.

The KOLBI KG500 Cassava Grater had an average Output Capacity of 694.8 kg/h of grated cassava at grater shaft speed of 1968 rpm

The noise emitted by the machine at 50 mm away from the operator's ear level was 88.2 db (A). The Fuel Consumption Rate of the 4.60 kW MITSUBISHI GB180 gasoline engine was 0.61 L/h.

Three persons were needed to operate the machine: one for hauling the cassava; one for feeding; and, one for collecting the grated cassava.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted at the plant site of TROPICS AGRO-INDUSTRIES, INC. in Naga City on May 11, 2013. The test was carried out to assess and verify the following:

1. Machine specifications
2. Performance
 - a. Output capacity
 - b. Speed of components
 - c. Noise level
 - d. Fuel Consumption
3. Labor requirement

OPERATOR'S MANUAL

An Operation and Instruction Manual was submitted which contains the following: Application, Features, Operation, Safety Instruction, and, Maintenance.

DESCRIPTION OF THE MACHINE

The KOLBI KG500 Cassava Grater as shown in **Figure 1**, was portable with circumferential-type grating unit. It was powered by a 4.6 kW MITSUBISHI GB180 gasoline engine which directly drove the grater's shaft through a coupler (**Fig. 2**). The machine consisted of the hopper and the grating chamber.

The hopper (**Fig. 3**) was where the materials to be grated were fed. It was made of G. I. sheet, Sch. #20 and installed above the grating chamber.

The grating chamber consisted of a circumferential-type grating cylinder (**Fig. 4a**) oriented horizontally with 4 mm embossed perforations which served as the blades of the grater. Two adjustable counter blades **Fig. 4b**) with dimensions (L x W x T) 310 mm x 32 mm x 6 mm were installed at the two opposite sides of the cylinder. Below the grating chamber was the outlet chute (**Fig. 5**) where grated cassava flow down.

The whole frame of the machine was made of 38 mm x 38 mm x 5 mm angle bar.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. KOLBI Cassava Grater
Model: KG500

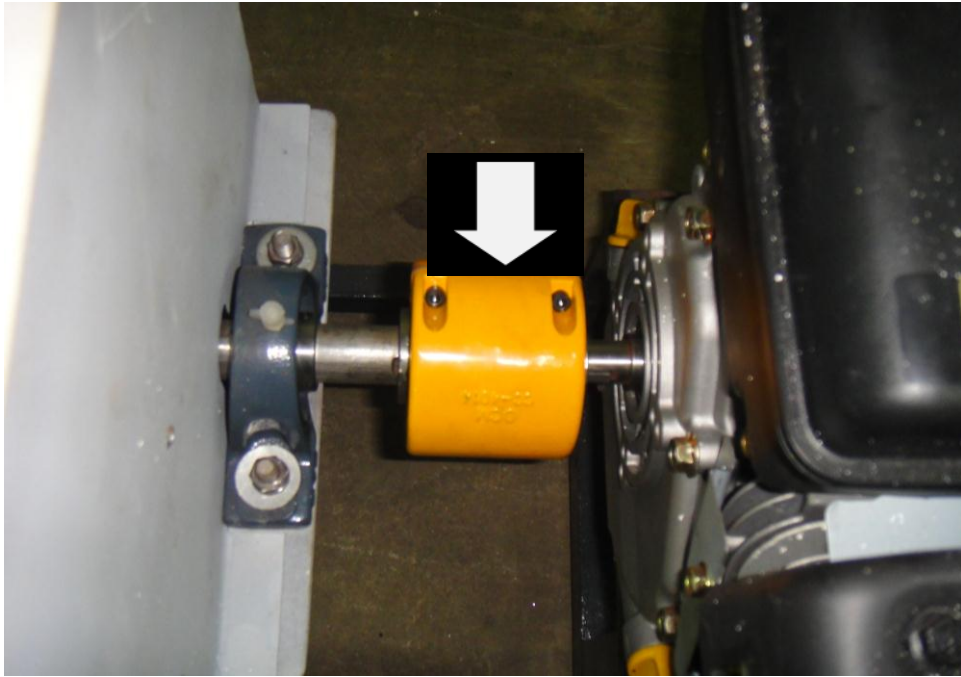


Figure 2. Coupler connecting the engine and the grater shaft



Figure 3. Hopper

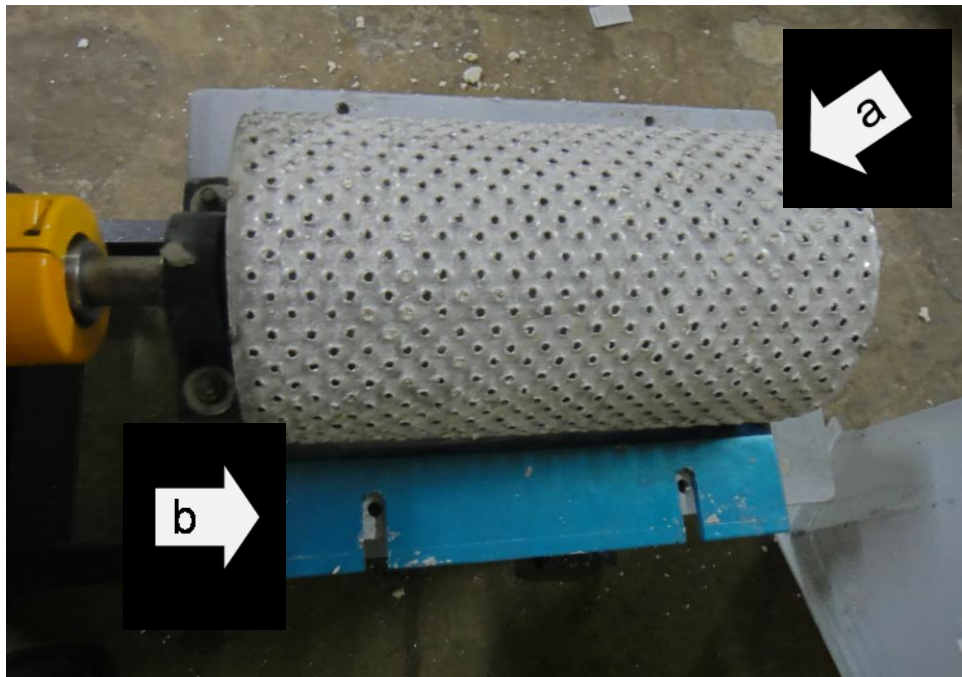


Figure 4. a) Grater cylinder
b) Counter blade



Figure 5. Outlet chute

SPECIFICATIONS

Table 1. Detailed specifications of the KOLBI KG500 Cassava Grater

A.	Brand	:	KOLBI
B.	Model	:	KG500
C.	Manufacturer	:	TROPICS AGRO INDUSTRIES, INC. 25 Panganiban Drive Naga City
D.	Overall Dimensions and Weight		
1.	Length (mm)	:	695
2.	Width (mm)	:	350
3.	Height (mm)	:	670
4.	Weight (kg) without engine	:	No Data
E.	Hopper		
1.	Inlet opening, L x W (mm)	:	300 x 125
2.	Height from the ground (mm)	:	670
3.	Material	:	G. I. Sheet, Ga. #20
F.	Grating Cylinder		
1.	Shape	:	Cylindrical
2.	Orientation	:	Horizontal
3.	Size, Length x Diameter (mm)	:	300 x 170
4.	Size of perforations (mm)	:	4

Table 1. Detailed specifications of the KOLBI KG500 Cassava Grater (cont'n.)

5.	Material	:	Stainless, #304
G.	Counter Blade		
1.	No. of pieces	:	2
2.	Size, L x W x T (mm)	:	310 x 32 x 6
3.	Material	:	Mild steel
H.	Discharge Chute		
1.	Size, L x W (mm)	:	300 x 125
2.	Height from the ground (mm)	:	75
3.	Material	:	G. I. Sheet, Ga. #20
I.	Main Frame		
1.	Material	:	Angle bar (38 mm x 38 mm x 5 mm)
J.	Engine (Prime mover)		
1.	Brand	:	MITSUBISHI
2.	Model	:	GB180
3.	Make	:	Japan
4.	Type	:	Four stroke, air-cooled single cylinder gasoline engine

Table 1. Detailed specifications of the KOLBI KG500 Cassava Grater (cont'n.)

5.	Rated Maximum Power (kW)	:	4.6
6.	Rated Speed (rpm)	:	3600
7.	Displacement (cc)	:	181
8.	Cooling system	:	Air-cooled
9.	Starting system	:	Rope recoil
10.	Dry weight (kg)	:	15.5

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 2-3**.

Output Capacity

Test results showed that using peeled cassava, the machine had an Output Capacity of 694.8 kg/h of grated cassava at grater cylinder shaft speed of 1968 rpm.

Noise Level

The noise emitted by the machine 50 mm away from the operator's ear level was 88.2 db(A).

Fuel Consumption

The 4.6 kW MITSUBISHI GB180 gasoline engine had a Fuel Consumption Rate of 0.61 L/h.

Labor Requirement

Three (**Fig. 6**) persons were needed to operate the machine: one for hauling the cassava, one for feeding, and, one for collecting the grated cassava.

Observations:

1. Peeled and manually cut cassava (**Figure 7**) was used in the test. The grater produced totally grated cassava (**Figure 8**).



Figure 6. Operators of the machine



Figure 7. Peeled cassava

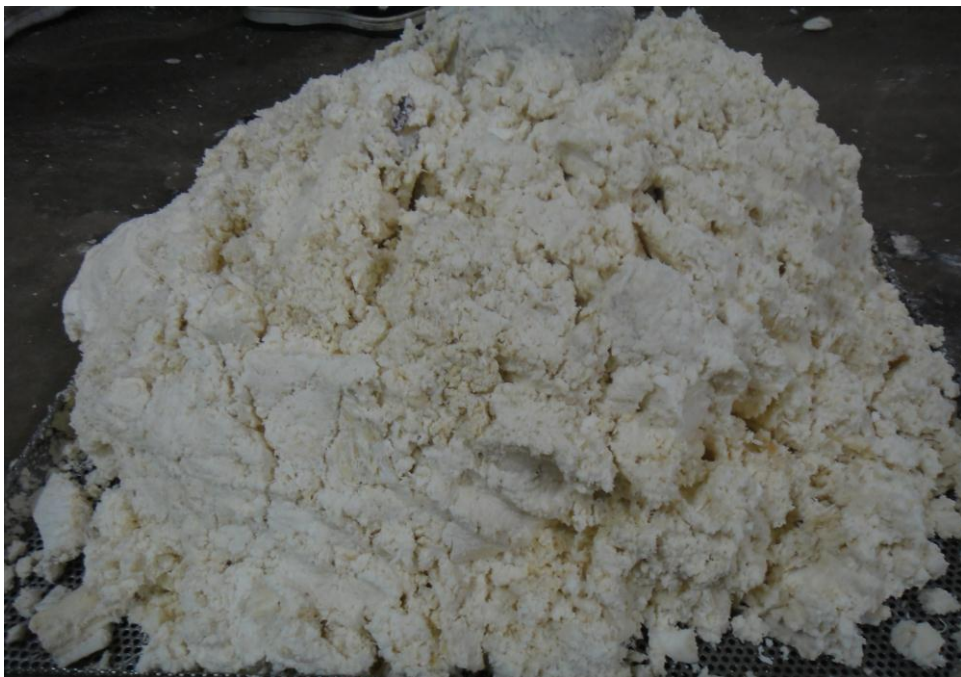


Figure 8. Grated cassava

PERFORMANCE TEST

Machine Tested : KOLBI Cassava Grater
Model: KG500

Date of Test : May 11, 2013

Location of Test : Naga City, Camarines Sur

Table 2. Conditions of Test Materials

ITEMS	Data
1. Crop	Cassava
2. Variety	“Lakan”
3. Moisture Content, $MC_{w.b.}$ (%)	54.2
4. Dimension, L x D (mm)	81 x 47
5. Condition	Peeled and manually cut

Table 3. Results of Performance Test

ITEMS	Data
1. Speed of Components	
a) Engine/Grater Shaft (rpm)	
Without load	2079
With load	1968
2. Noise Level, [db(A)]	
Without load	82.4
With load	88.2
3. Prime mover's Fuel Consumption (MITSUBISHI GB180), L/h	0.61
4. Output Capacity, kg/h	694.8

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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